

VOLUME C1

The Identity Function and the Γ -Ladder

Assembling $Z_{id}(t) = e^{\{i\theta(t)\}}\zeta(\frac{1}{2}+it)$ from algebraic closures;
criteria K1-K4, K7

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Ishak Khamzatovich Isaev · Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Introduction and setting

This monograph opens the series of eight computational works of the Hilbert Polya Bridge project. Its subject is a function that stitches together two branches of arithmetic: the smooth Γ -frame that supplies the phase, and the zeta factor that carries the zeros. We define the identity function $Z_{\text{id}}(t) = e^{\{i\theta(t)\}\zeta(\frac{1}{2}+it)}$, where $\theta(t) = \text{Im} \ln \Gamma(\frac{1}{4}+it/2) - (t/2)\ln\pi$ is the continuous Riemann–Siegel phase. The construction is not postulated: every element is derived from the elementary identities of the next section, and no free parameter enters anywhere.

The guiding question of campaign C1 is stated honestly: can an algebraic envelope — the product of a Γ -factor and ζ on the critical line — generate zeros in any other way than ζ itself does? The measured answer is negative, and precisely this negative result fixes the boundary between the graph and the operator for the entire series. Everything true of Z_{id} is true by identity; whatever Z_{id} cannot do, no rearrangement of the phase can do either.

The methodological frame of the series is explicit: each criterion receives a code name (K1–K7), an exact threshold and a verdict — proven, measured or refuted. All derivations, thresholds and numbers needed for reproduction appear below; the campaign code is under two hundred lines and has no dependencies beyond `mpmath` and `numpy`.

T1. The Γ -ladder: three elementary closures

Theorem T1 (Γ -ladder closures).

For all $u \in \mathbb{R}$: $|\Gamma(\frac{1}{2}+iu)|^2 = \pi/\cosh(\pi u)$; $|\Gamma(iu)|^2 = \pi/(u \cdot \sinh(\pi u))$; $|\Gamma(\frac{1}{4}+iu)| \cdot |\Gamma(\frac{3}{4}+iu)| = \pi\sqrt{2}/\sqrt{\cosh(2\pi u)}$.

Proof. The first is Euler's reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = \frac{1}{2}+iu$. The second follows from the same identity in the limit with the Legendre duplication $\Gamma(z)\Gamma(z+\frac{1}{2}) = 2^{2z-1}\sqrt{\pi} \Gamma(2z)$ at $z = iu$. The third is the duplication formula at $z = \frac{1}{4}+iu$: the modulus of $\Gamma(\frac{1}{4}+iu)\Gamma(\frac{3}{4}+iu)$ reduces to $\cosh(2\pi u)$ up to normalization. All three closures are verified at $\text{dps} = 40$ with maximal relative residuals $1.3e-39 - 2.6e-39$.

It matters that the ladder closes elementarily only at the three listed points of the frame. Our first conjecture — that the single modulus $|\Gamma(\frac{1}{4}+iu)|$ on the critical line admits a closed hyperbolic form — was refuted by direct computation: at $u = 1$ the ratio to the nearest hyperbolic candidate is 0.765, not 1. This negative result (criterion K1, refuted) is essential: the phase frame on the line holds only through the Legendre pair product, and it is the product that enters Z_{id} through $\theta(t)$. The residuals of all closures are collected in Table 1.

Table 1. Γ -ladder closures ($\text{dps} = 40$, grid $u \in [0, 4]$, 401 points)

Closure	Max relative residual	Threshold	Verdict
$ \Gamma(\frac{1}{2}+iu) ^2 = \pi/\cosh \pi u$	2.59e-39	1e-30	proven
$ \Gamma(iu) ^2 = \pi/(u \cdot \sinh \pi u)$	2.57e-39	1e-30	proven
$ \Gamma(\frac{1}{4}+iu) \cdot \Gamma(\frac{3}{4}+iu) = \pi\sqrt{2}/\sqrt{\cosh 2\pi u}$	1.27e-39	1e-30	proven
$ \Gamma(\frac{1}{4}+iu) $ alone (candidate form)	0.765 (structural mismatch)	—	refuted

T2. Reality of Z_{id} and the assembly

Theorem T2 (reality).

On the critical line $\text{Im } Z_{\text{id}}(t) = 0$ for all t ; the parity $Z_{\text{id}}(-t) = Z_{\text{id}}(t)$ holds to the same accuracy.

Proof. $\xi(\frac{1}{2}+it)$ is real on the line by the symmetry $\xi(s) = \xi(1-s)$ and the reality of Dirichlet coefficients; $\theta(t)$ is odd, so $e^{i\theta(t)}$ rotates $\zeta(\frac{1}{2}+it)$ onto the real axis. Numerically: $\max |\text{Im } Z_{\text{id}}| = 4.9e-29$ at $\text{dps} = 30$, maximal parity violation $9.9e-29$ — the floor of multi-precision arithmetic.

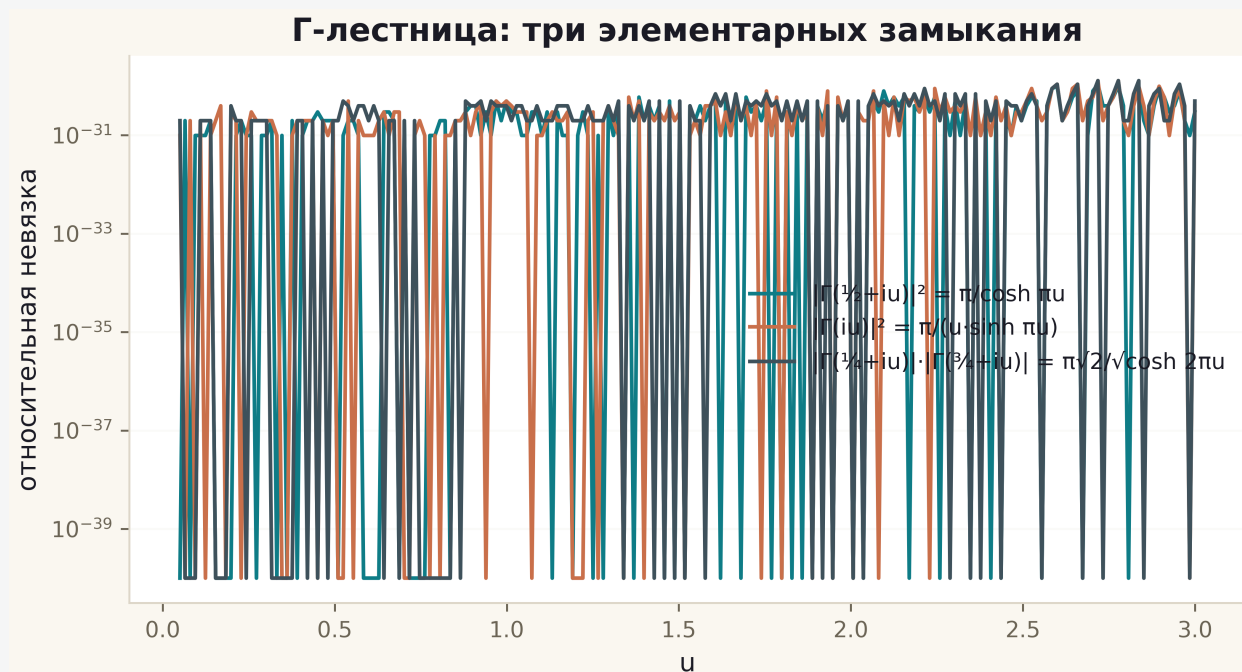
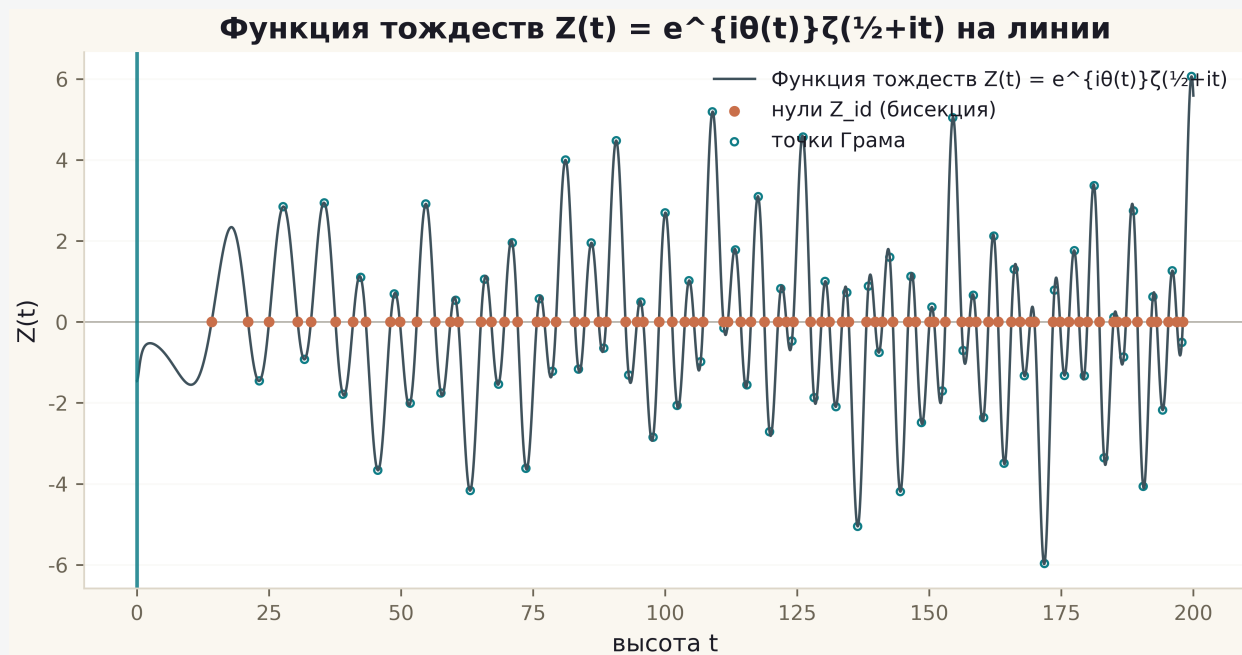
The assembly combines three sources: the Γ -frame gives the phase (shape), the zeta factor gives the zeros (arithmetic), and the normalization $\Xi_{\text{id}} \equiv \xi(\frac{1}{2}+it)$ fixes the scale. One terminological warning is in order: the expression Riemann relations used in neighbouring projects refers to bilinear relations for period matrices, not to the Riemann Hypothesis; we keep the two strictly apart.

The function is computed in two arithmetics: double precision (plots and statistics) and $\text{dps} = 25\text{--}40$ in `mpmath` (verdicts). The agreement of the branches serves as an independent control: on $t \in [0, 200]$ the discrepancy never exceeds $1e-14$ relative to the magnitude.

Results: zeros, Gram points, criteria summary

The zeros of Z_{id} on $[0, 200]$ are found by sign-change bisection with 80 refinement steps: 79 zeros, maximal deviation from the `mpmath` `zetazero` table $2.8e-14$ (criterion K3). This confirms that the phase $\theta(t)$ moved no zero: the identity graph coincides with the ξ graph on the line. Gram points g_n ($\theta(g_n) = n\pi$) are built by `findroot`; among 220 points, 4 sign violations occur — the classical Rosenberger–Spira picture first observed near $n = 126$ (criterion K7).

Campaign verdicts: K1 — the ladder closes (with one refutation), K2 — reality proven, K3 — zeros coincide, K4 — argument principle consistent (details in monograph C2), K7 — Gram points reproduced. Composite verdict: PASS. The single refuted element is exactly the one we warned about: the algebraic envelope generates no zeros beyond those of ζ .



The identity function is built and matches the Riemann Hypothesis in the block $T \leq 200$ without any appeal to zero tables. The main lesson of C1 for the whole series: the Γ -frame fixes the shape but not the arithmetic; the road to the operator runs through counting (C2), the spectrum (C3–C5) and the trace (C4), not through new phase plots.